

What Survives the Relaxation

Four boundaries, from amnesia to agency

A SOAP FILM blown into a bubble finds its sphere without being told anything about how it was blown. Surface tension pulls the film downhill in *free energy*, the portion of a system's energy actually available to do work, until the enclosed volume is wrapped in the least possible area, and the configuration it lands on is set entirely by the ambient physics of the medium, which is why every bubble of a given volume is the same bubble; nothing about the operator, the wand, the breath, or the irregular blob the film passed through on its way survives into the final shape. The bubble is a boundary with no memory. That erasure is not a defect of soap but the condition of its convergence, since a process that preserved its initial conditions would have no way of arriving at one answer from many beginnings.

Archimedes converged on the circle by the opposite route, doubling the sides of inscribed and circumscribed polygons until the bounds closed, and every refinement preserved what came before. A bubble converges by discarding. This story follows the second route.

The decisive complication comes when one enclosure becomes many. Partition is another matter. Hales showed that the hexagon beats every equal-area partition of the plane, irregular mixtures included; in three dimensions Kelvin's foam held the record for a century before Weaire and Phelan beat it with two distinct cell shapes in a repeating group of eight. The plane is unusually generous in allowing a single regular tile to win at all. Irregularity is not something biology introduces into a world of Platonic forms. It is already waiting in the foam.

The honeycomb is a two-dimensional partition with a third dimension along which nothing much is asked, and it is the first case where something about the occupant survives. Every cell is asked the same question, so the walls repeat the same answer. The bees begin the cells; warm wax, softened by the colony and drawn by surface tension, finishes the calculation, meeting at the angles a soap film would have chosen, which makes the comb a relaxation of the same *dissipative* kind as the bubble, one that sheds energy to its surroundings on the way down. What the relaxation cannot erase is the width. Wax economy is scale-free and would happily build fewer, larger cells to improve the wall-to-volume ratio, so the size is fixed instead by the girth of the larva that has to fit inside. One measurement of the occupant survives, and nothing else about it does.

At the far end, the boundary remembers in detail. *Protein folding* is the process by which a linear chain of amino acids settles into a three-dimensional shape, another downhill relaxation, driven here by *hydrophobic burial*, the sheltering of water-repelling parts of the chain in the interior, together with hydrogen bonding, rather than by surface tension. Yet it converges towards one highly specific irregular structure instead of a universal symmetric one. The sequence has written so much into the *energy landscape* being descended, the map of which configurations lie lower than which, that the dissipative dynamics develops that information rather than washing it out. The resulting surface does a different job at nearly every point, which is why it looks nothing like a hexagon. Irregularity here is not a failure to achieve the comb's order; it is what an optimum looks like when the demands on a boundary differ from place to place. Biology supplies its own control, since the structural proteins that face similar demands along their whole length, collagen and keratin and silk fibroin, form periodic repeating structures, while the enzymes and receptors that must recognise particular molecular partners fold into idiosyncrasy.

Yet the sequence is not a clean specification of the task the protein performs now. It is a palimpsest of workable alterations: *residues*, the

individual amino-acid units of the chain, demanded by present function; residues inherited from circumstances that have since changed; and residues that persist because removing them brought no advantage. Evolution cannot return to a blank chain, since it edits a molecule already in service, and a contingent alteration can therefore become the foundation on which later necessities are built. A feature that has lost its original reason may have become load-bearing in the meantime, yesterday's happenstance hardening into today's constraint. The folded surface preserves more than purpose. Some of its irregularity is instruction, some compromise, some neutral history, and some legacy whose original reason has disappeared.

Set that against the bubble and physics has a name for the contrast. In classical general relativity the *no-hair theorem* holds that settled black holes of equal mass, charge and angular momentum are indistinguishable from outside, however different the stars that collapsed into them, and the same austerity operates at incomparably gentler stakes whenever a dissipative equilibrium keeps only the parameters its dynamics require, which is why two bubbles of equal volume are indistinguishable however differently they were blown. Sphericity is what a boundary looks like when nothing contingent has been allowed to survive its own formation. The protein refuses that amnesia, and its irregular surface is the visible residue of everything the bubble discards.

Memory is still not agency. A surface that admits one molecule and turns away the rest looks like Maxwell's demon, who watches individual particles and opens his door according to what he sees. The comparison fails on the watching. A *binding pocket*, the cleft in a folded protein that a particular partner fits into, sees nothing and learns nothing; its selectivity was installed once by the sequence and is thereafter passive, a filter with pre-shaped holes rather than a gatekeeper making judgements. The specificity was paid for upstream, in the synthesis of the chain and much further upstream in the selection of the sequence that specifies it.

The discontinuity lies one step further out, in the molecular machines that consume free energy every cycle to rectify thermal noise: kinesin walking directionally along a cellular track, the ribosome rejecting a mispaired transfer RNA, ATP synthase turning a proton gradient into chemical work. The ribosome's mechanism is worth naming, since in *kinetic proofreading* the consumption of fuel buys a second opportunity to reject a near match, converting a slight equilibrium preference into reliable discrimination. The contrast is between a relaxed structure whose selectivity is already embodied in its shape, a shape that has finished falling and will stay where it landed, and a driven steady state whose organised activity stops when its energy supply does. Flux alone does not account for accumulated organisation, since a flame is driven and a convection cell is ordered and neither founds a lineage. Flux supplies the work, heredity supplies retention, and selection supplies the ratchet, which is why the machine's structure could accumulate across generations while the flame's cannot.

Relaxation itself never changes its behaviour through any of this. It always runs downhill. What changes is where the information has been put: in the bubble it lies in the initial shape, so the fall destroys it; in the protein it lies in the sequence-defined landscape, so the same fall develops it like a photograph; in the machine, the fall of a proton gradient is coupled to a motion that would otherwise be improbable. Medium, occupant, inheritance, power. At the last step a surface stops being a solution to a problem and becomes a mechanism for solving one.